

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 - Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	K. Madhurmitha	Reg. No.:	192513069
Section / Batch:	BE-EEE (2 nd year)	Date:	20/07/2026

Code of Conduct

I K. Madhurmitha 192513069 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Madh

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

Name : K. Madhumidha
Reg No : 192513069
Course : Data structure
Course code : CSA0305
Date : 5/08/26

- 1) Find the differences between the minimum and maximum values in a single linked list having

34 → 77 → 22 → 55 → 66 → 11 → in the list

AIM :

The aim of this program is to create a doubly linked list, display all elements in both forward and reverse directions

PROBLEM :

A single linked list contains the element 34 → 77 → 22 → 55 → 66 → 11. Find the minimum value, maximum value, and calculate the difference between them.

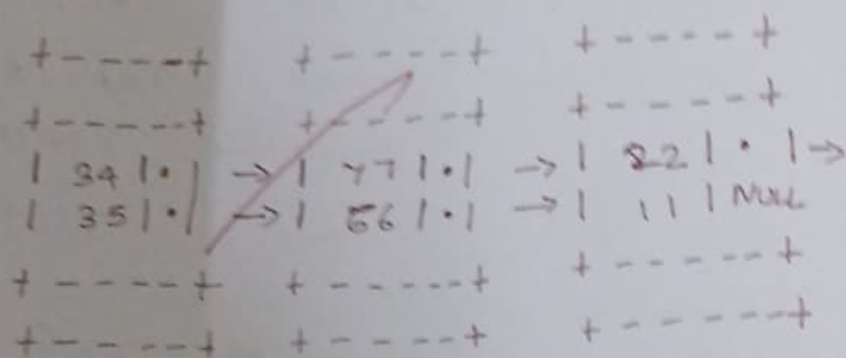
DATA STRUCTURE USED :

A ~~doubly~~ ^{single} linked list is a linear data structure where each node contains :

- * Previous pointer (prev)
- * Data
- * Next pointer

DIAGRAM:

Head



PROCESS:

- * The process starts by creating a singly linked list.
- * The first node is assumed to be both the minimum
- * $\text{let min} = 34$ and $\text{max} = 34$
- * Compare 77:
 $\rightarrow 77 > 34 \rightarrow \text{min} = 22$
- * compare 22:
 $\rightarrow 22 < 34 \rightarrow \text{min} = 22$
- * Compare 55:
 $\rightarrow \text{No change}$
- * Compare 66:
 $\rightarrow \text{No change}$
- * Compare 11:
 $\rightarrow 11 < 22 \rightarrow \text{min} = 11$
- * End of the list
- * Calculate
 $\rightarrow \text{difference} = \text{max} - \text{min} = 77 - 11 = 66$

* Display the result

Output:

* Minimum = 11

* Maximum = 77

* Difference = 66

Dry Run

Node value	Minimum	Maximum
34	34	34
77	34	77
22	22	77
55	22	77
66	22	77
11	11	77

Time Complexity

+ $O(n)$

Space complexity

+ $O(1)$

Result

Difference between minimum and maximum values = 66

- 2) The double linked list has two links in each node as "prev", "data" and "next". Display the elements from DLL in forward and reverse order and ~~filter~~ find the average of those 'n' elements

AIM:

To display the elements of a doubly linked list in forward and reverse order and ~~calculations~~ the average of all elements.

PROBLEM:

Given the doubly linked: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50$.

DISPLAY:

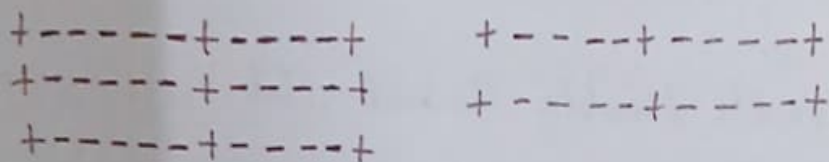
- * Forward traversal
- * Reverse traversal
- * ~~Average~~ of all elements

DATA STRUCTURE USED:

- * Doubly linked list
- * Each node contains:
 - i) Previous pointer (prev)
 - ii) Data
 - iii) Next pointer (next)

DIAGRAM

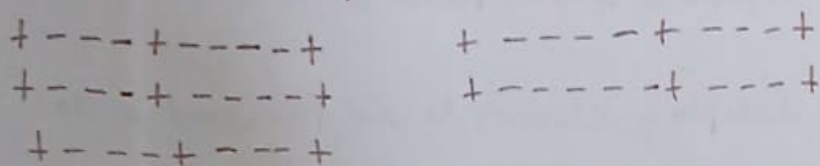
NULL



| Prev | 10 | Next | $\leftarrow$ - $\rightarrow$ | Prev | 20 |

Next | $\leftarrow$ - $\rightarrow$ | Prev | 30 | Next | $\leftarrow$ - $\rightarrow$ | Prev |

40 | Next | $\leftarrow$ - $\rightarrow$ | Prev | 50 | Next |



|

NULL

Forward Traversal

Head



10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ 40 $\rightarrow$ 50

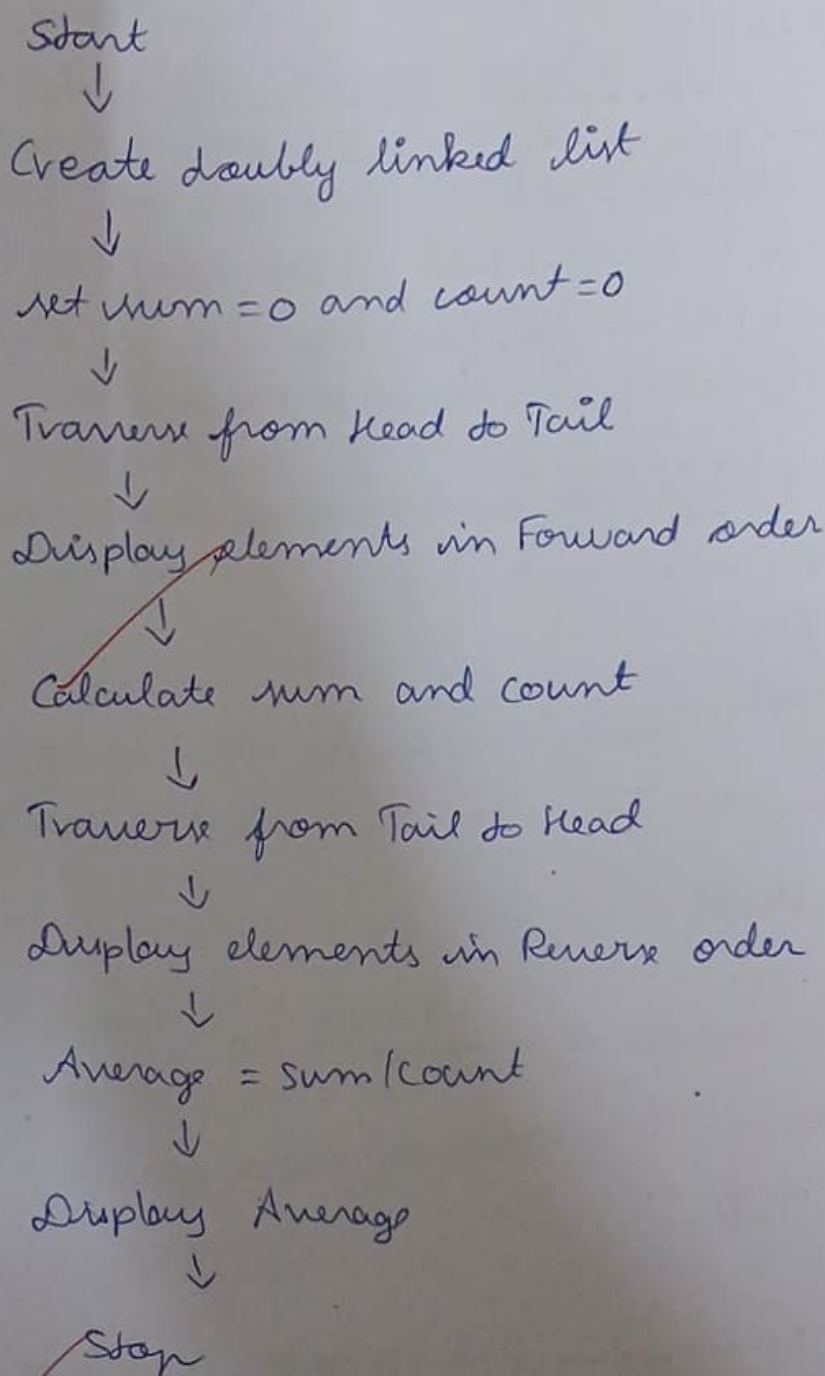
Reverse Traversal

Tail



50 $\rightarrow$ 40 $\rightarrow$ 30 $\rightarrow$ 20 $\rightarrow$ 10

PROCESS :



Example :

List : 10 → 20 → 30 → 40 → 50

Forward Traversal :

10 → 20 → 30 → 40 → 50

Reverse Traversal

50 $\rightarrow$ 40 $\rightarrow$ 30 $\rightarrow$ 20 $\rightarrow$ 10

CALCULATION:

Element	sum
10	10
20	30
30	60
40	100
50	150

Number of elements = 5

Average = $150 / 5 = 30$

Time complexity

* Forward Traversal = $O(n)$

* Reverse Traversal = $O(n)$

* Overall = $O(n)$

Space complexity

* $O(1)$

RESULT :

- * Forward display : 10 20 30 40 50
- * Reverse display : 50 40 30 20 10
- * Average : 30.0

Ugita